*Table 12.* **Ablation study.** Trained on TinyImageNet with 1-step PGD and evaluated under 100-step PGD, CW, and AutoAttack (AA) with $\epsilon = 1/255$. Results are averaged over 16 datasets. Best and second-best are in **bold** and underline.

| Methods | Clean | PGD | CW | AA |
|---|---|---|---|---|
| CLIP | 64.45 | 0.05 | 4.06 | 0.51 |
| $\mathcal{L}_{\text{ce}}$ | 43.83 | 29.86 | 29.25 | 28.74 |
| $\mathcal{L}_{\text{ce}}$ + KL$(p(x)\|p(x^a))$ | 45.03 | 30.12 | 29.61 | 29.13 |
| $\mathcal{L}_{\text{ce}}$ + KL$(p(x^a)\|p(x))$ | 45.05 | 29.98 | 29.28 | 28.80 |
| $\mathcal{L}_{\text{ce}}$ + KL$(\text{Dir}(\alpha\|\alpha_{adv}))$ | 36.72 | 25.01 | 24.66 | 24.36 |
| $\mathcal{L}_{\text{ce}}$ + KL$(\text{Dir}(\alpha_{adv}\|\alpha))$ | **54.17** | **32.20** | **31.41** | **30.58** |

*Table 13.* **Zero-shot adversarial robustness across 16 single-label datasets.** All methods are fine-tuned on TinyImageNet following TGA-ZSR (Yu et al., 2024), adversarial training uses 2-step PGD (Madry et al., 2017) with $\epsilon = 1/255$. *Average* is the mean across datasets. $H$ is the harmonic mean between Clean and the corresponding robust score. Best and second-best are in **bold** and underline.

| | Methods | TinyImageNet | Cifar10 | Cifar100 | STL10 | SUN397 | Food101 | Oxfordpets | Flowers102 | DTD | EuroSAT | FGVC Aircraft | ImageNet | Caltech101 | Caltech256 | StanfordCars | PCAM | Average | H |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Clean | CLIP (Radford et al., 2021) | 57.96 | 88.03 | 60.45 | 97.03 | 57.26 | 83.89 | 87.41 | 65.49 | 40.64 | 42.66 | 20.16 | 59.15 | 85.32 | 81.73 | 52.02 | 52.08 | 64.45 | |
| | TeCoA (Mao et al., 2022) | 71.24 | 67.56 | 38.26 | 85.89 | 36.01 | 28.23 | 61.30 | 32.04 | 24.95 | 16.13 | 5.19 | 32.89 | 72.16 | 59.00 | 20.28 | 50.11 | 43.83 | |
| | FARE (Schlarmann et al., 2024) | 41.86 | 79.81 | 48.27 | **94.24** | 46.15 | **58.90** | **80.98** | 47.63 | 23.09 | **24.19** | **15.63** | 42.93 | 78.22 | 72.05 | **43.96** | 50.02 | 53.00 | |
| | PMG-AFT (Wang et al., 2024) | 48.60 | 74.73 | 43.59 | 90.41 | **51.70** | 56.52 | 79.40 | **48.43** | **32.45** | 21.76 | 11.79 | **46.74** | **82.49** | **73.59** | 41.21 | **56.13** | 53.72 | |
| | TGA-ZSR (Yu et al., 2024) | **76.60** | 79.18 | 47.37 | 90.65 | 43.10 | 38.90 | 68.44 | 39.81 | 25.69 | 19.70 | 8.82 | 39.27 | 76.42 | 66.31 | 28.44 | 49.92 | 49.91 | |
| | Comp-TGA (Yu et al., 2026) | 75.00 | **81.91** | 50.67 | 90.84 | 46.45 | 44.28 | 72.04 | 41.41 | 28.35 | 23.50 | 9.12 | 42.23 | 78.96 | 69.25 | 29.55 | 49.88 | 52.09 | |
| | UCAT (Ours) | 74.46 | 81.81 | **54.45** | 91.88 | 41.06 | 53.58 | 74.16 | 47.57 | 31.92 | 19.29 | 10.95 | 43.20 | 82.39 | 71.53 | 37.32 | 51.20 | **54.17** | |
| PGD | CLIP (Radford et al., 2021) | 0.19 | 9.57 | 3.07 | 23.64 | 0.62 | 0.34 | 0.64 | 1.62 | 2.22 | 0.00 | 0.00 | 0.48 | 5.65 | 7.19 | 0.02 | 0.06 | 3.46 | 6.56 |
| | TeCoA (Mao et al., 2022) | 50.96 | 39.33 | 21.64 | 69.78 | 20.07 | 13.50 | 37.80 | 19.17 | 18.30 | **11.88** | 2.16 | 18.47 | 56.00 | 42.38 | 9.33 | 46.92 | 29.86 | 35.52 |
| | FARE (Schlarmann et al., 2024) | 3.78 | 7.83 | 2.80 | 48.18 | 5.66 | 2.45 | 10.93 | 6.52 | 5.75 | 0.08 | 0.54 | 5.20 | 33.21 | 20.70 | 2.31 | 48.97 | 12.81 | 20.63 |
| | PMG-AFT (Wang et al., 2024) | 19.18 | **51.39** | 27.23 | 72.63 | 20.05 | 16.88 | 44.59 | 26.43 | **20.05** | 11.49 | 3.21 | 18.09 | 61.13 | 43.46 | 14.80 | **55.52** | 31.63 | 39.82 |
| | TGA-ZSR (Yu et al., 2024) | **50.68** | 42.16 | 22.82 | 72.18 | 21.57 | 16.53 | 39.96 | 22.44 | 17.82 | 11.75 | 2.88 | 20.39 | 58.05 | 46.18 | 11.40 | 48.05 | 31.55 | 38.66 |
| | Comp-TGA (Yu et al., 2026) | 49.90 | 40.44 | 22.43 | 72.56 | **21.61** | 16.81 | 41.56 | 23.04 | 18.88 | 11.66 | 2.70 | 20.07 | 58.80 | 46.42 | 10.50 | 44.98 | 31.40 | 39.18 |
| | UCAT (Ours) | 47.56 | 43.81 | **25.16** | **73.83** | 20.44 | **22.86** | **45.11** | **26.79** | 19.47 | 2.99 | **3.45** | **22.22** | **65.32** | **50.47** | **15.30** | 30.37 | **32.20** | **40.39** |
| CW | CLIP (Radford et al., 2021) | 0.14 | 9.91 | 3.34 | 26.01 | 1.16 | 0.51 | 0.87 | 2.03 | 2.55 | 0.01 | 0.00 | 1.10 | 6.82 | 8.17 | 2.32 | 0.04 | 4.06 | 7.64 |
| | TeCoA (Mao et al., 2022) | 50.16 | 38.62 | 20.76 | 69.55 | 18.84 | 12.46 | 37.37 | 18.12 | 17.23 | **11.63** | 2.10 | 17.70 | 55.62 | 41.70 | 9.23 | 46.88 | 29.25 | 35.08 |
| | FARE (Schlarmann et al., 2024) | 4.10 | 4.12 | 2.96 | 43.35 | 6.07 | 3.17 | 15.15 | 5.66 | 4.52 | 0.12 | 1.11 | 5.34 | 32.50 | 20.85 | 4.38 | 48.86 | 12.64 | 20.41 |
| | PMG-AFT (Wang et al., 2024) | 13.16 | 42.10 | 21.31 | 65.69 | 13.12 | 11.43 | 28.05 | 17.53 | 12.55 | 8.51 | 0.99 | 11.72 | 52.84 | 35.68 | 7.06 | 14.26 | 22.25 | 31.47 |
| | TGA-ZSR (Yu et al., 2024) | **50.80** | 42.24 | 22.64 | 71.99 | 20.83 | 16.03 | 40.20 | 21.52 | 16.97 | 11.56 | 2.85 | 20.01 | 57.72 | 45.84 | 11.23 | **48.03** | 31.28 | 38.46 |
| | Comp-TGA (Yu et al., 2026) | 49.82 | 40.40 | 22.03 | 72.33 | **21.13** | 15.95 | 42.25 | 22.23 | 17.55 | 11.41 | 2.64 | 19.94 | 58.66 | 46.13 | 11.09 | 45.01 | 31.16 | 38.99 |
| | UCAT (Ours) | 47.08 | **43.30** | **23.92** | **73.55** | 19.20 | **21.68** | **45.38** | **24.95** | **17.87** | 2.41 | **3.21** | **21.14** | **64.63** | **49.54** | **14.75** | 29.89 | **31.41** | **39.76** |
| Auto Attack | CLIP (Radford et al., 2021) | 0.00 | 2.54 | 1.11 | 3.18 | 0.05 | 0.03 | 0.03 | 0.02 | 0.19 | 0.17 | 0.23 | 0.04 | 0.10 | 0.26 | 0.07 | 0.12 | 0.51 | 1.01 |
| | TeCoA (Mao et al., 2022) | **49.44** | 37.87 | 20.45 | 69.31 | 17.41 | 12.19 | 36.58 | 17.81 | 17.29 | **11.42** | 1.86 | 17.19 | 54.95 | 41.19 | 8.16 | 46.79 | 28.74 | 34.72 |
| | FARE (Schlarmann et al., 2024) | 0.12 | 0.03 | 0.21 | 10.18 | 0.84 | 0.19 | 0.93 | 0.60 | 1.92 | 0.07 | 0.06 | 0.86 | 10.26 | 5.59 | 0.21 | 5.15 | 2.33 | 4.45 |
| | PMG-AFT (Wang et al., 2024) | 8.22 | 41.86 | 21.18 | 65.45 | 7.95 | 7.34 | 18.94 | 12.59 | 3.13 | 7.17 | 0.51 | 7.90 | 44.91 | 28.29 | 3.22 | 7.41 | 17.88 | 26.83 |
| | TGA-ZSR (Yu et al., 2024) | 49.26 | 40.92 | 21.75 | 71.55 | **19.88** | 15.32 | 38.84 | 20.98 | 17.02 | 11.26 | 2.34 | 19.12 | 57.11 | 45.16 | 9.87 | **48.00** | 30.52 | 37.88 |
| | Comp-TGA (Yu et al., 2026) | 39.84 | 30.11 | 17.23 | 65.48 | 15.70 | 12.50 | 34.10 | 18.21 | 14.84 | 10.82 | 1.74 | 15.87 | 52.61 | 40.69 | 7.38 | 42.81 | 26.24 | 34.90 |
| | UCAT (Ours) | 45.80 | **42.32** | **23.03** | **73.15** | 18.26 | **20.52** | **44.02** | **24.54** | **18.14** | 2.26 | **2.61** | **20.15** | **63.73** | **48.66** | **12.60** | 29.51 | **30.58** | **39.09** |

*Table 14.* **Zero-shot adversarial robustness on multi-label dataset MS-COCO (Lin et al., 2014).** All ZSAR models are adversarially trained on TinyImageNet with the FARE (Schlarmann et al., 2024) 10-step PGD setting ($\epsilon = 2/255$), and evaluated under CW-100 at radii $\epsilon \in \{1/255, 2/255, 4/255\}$ plus clean. We report mean Average Precision (mAP), Precision (P), Recall (R), and F1-score (F1) at top-3 predictions. $H(\text{F1@3})$ denotes the harmonic mean of clean and adversarial F1@3. Best and second-best are in **bold** and underline.

| | Clean | | | | $\epsilon = 1/255$ | | | | | $\epsilon = 2/255$ | | | | | $\epsilon = 4/255$ | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Methods | mAP | P@3 | R@3 | F1@3 | mAP | P@3 | R@3 | F1@3 | H(F1@3) | mAP | P@3 | R@3 | F1@3 | H(F1@3) | mAP | P@3 | R@3 | F1@3 | H(F1@3) |
| CLIP (Radford et al., 2021) | 51.96 | 45.33 | 46.49 | 45.9 | 17.72 | 25.21 | 25.85 | 25.52 | 32.80 | 6.67 | 10.07 | 10.32 | 10.19 | 16.68 | 3.77 | 3.84 | 3.94 | 3.89 | 7.17 |
| TeCoA (Mao et al., 2022) | 46.34 | 33.73 | 34.57 | 34.14 | 37.32 | 30.23 | 30.99 | 30.60 | 32.27 | 27.94 | 24.85 | 25.48 | 25.16 | 28.97 | **17.63** | 19.67 | 20.17 | 19.91 | 25.15 |
| FARE (Schlarmann et al., 2024) | **49.21** | **43.83** | **44.95** | **44.37** | 29.18 | 33.45 | 34.30 | 33.86 | 38.41 | 16.18 | 22.44 | 23.02 | 22.72 | 30.05 | 8.12 | 13.21 | 13.55 | 13.38 | 20.56 |
| PMG-AFT (Wang et al., 2024) | 48.94 | 41.77 | 42.83 | 42.29 | 29.75 | 32.32 | 33.15 | 32.72 | 36.89 | 17.16 | 23.37 | 23.98 | 23.67 | 30.35 | 8.78 | 13.45 | 13.8 | 13.62 | 20.60 |
| TGA-ZSR (Yu et al., 2024) | 48.61 | 37.19 | 38.13 | 37.65 | **38.23** | 32.95 | 33.79 | 33.36 | 35.38 | 29.03 | 27.70 | 28.40 | 28.04 | 32.14 | 15.18 | 18.64 | 19.11 | 18.87 | 25.14 |
| Comp-TGA (Yu et al., 2026) | 44.44 | 37.62 | 38.57 | 38.08 | 30.82 | 28.66 | 29.39 | 29.01 | 32.93 | 25.51 | 25.48 | 26.13 | 25.80 | 30.76 | 17.79 | 20.65 | 21.18 | 20.91 | 27.00 |
| UCAT (Ours) | 47.14 | 41.55 | 42.62 | 42.07 | 37.60 | **36.58** | **37.52** | **37.04** | **39.40** | **29.33** | **31.39** | **32.20** | **31.78** | **36.21** | 14.61 | **21.33** | **21.86** | **21.59** | **28.54** |

*Table 15.* **Zero-shot adversarial robustness on multi-label dataset MS-COCO (Lin et al., 2014).** All ZSAR models are adversarially trained on TinyImageNet with the FARE (Schlarmann et al., 2024) 10-step PGD setting ($\epsilon = 2/255$), and evaluated under **PGD**-100 at radii $\epsilon \in \{1/255, 2/255, 4/255\}$ plus clean. We report mean Average Precision (mAP), Precision (P), Recall (R), and F1-score (F1) at top-3 predictions. $H(\text{F1@3})$ denotes the harmonic mean of clean and adversarial F1@3. Best and second-best are in **bold** and underline.

| | Clean | | | | $\epsilon = 1/255$ | | | | | $\epsilon = 2/255$ | | | | | $\epsilon = 4/255$ | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Methods | mAP | P@3 | R@3 | F1@3 | mAP | P@3 | R@3 | F1@3 | H(F1@3) | mAP | P@3 | R@3 | F1@3 | H(F1@3) | mAP | P@3 | R@3 | F1@3 | H(F1@3) |
| CLIP (Radford et al., 2021) | 51.96 | 45.33 | 46.49 | 45.9 | 40.21 | 40.43 | 41.45 | 40.93 | 43.27 | 24.94 | 27.85 | 28.54 | 28.19 | 34.93 | 11.30 | 15.55 | 15.94 | 15.74 | 23.44 |
| TeCoA (Mao et al., 2022) | 46.34 | 33.73 | 34.57 | 34.14 | 38.90 | 39.14 | 40.14 | 39.63 | 36.68 | 42.10 | 30.54 | 31.32 | 30.92 | 32.45 | **31.29** | 25.55 | 26.21 | 25.87 | 29.44 |
| FARE (Schlarmann et al., 2024) | **49.21** | **43.83** | **44.95** | **44.37** | 43.33 | 38.63 | 39.61 | 39.11 | **41.57** | 23.82 | 32.89 | 33.74 | 33.30 | 38.05 | 11.21 | 24.31 | 24.93 | 24.61 | 31.66 |
| PMG-AFT (Wang et al., 2024) | 48.94 | 41.77 | 42.83 | 42.29 | **46.42** | 36.00 | 36.91 | 36.44 | 39.15 | 34.56 | 32.91 | 33.75 | 33.32 | 37.27 | 15.51 | 25.16 | 25.81 | 25.47 | 31.79 |
| TGA-ZSR (Yu et al., 2024) | 48.61 | 37.19 | 38.13 | 37.65 | 44.62 | 32.33 | 33.15 | 32.73 | 35.02 | **42.79** | 33.52 | 34.37 | 33.93 | 35.69 | 29.20 | 27.31 | 28.01 | 27.65 | 31.88 |
| Comp-TGA (Yu et al., 2026) | 44.44 | 37.62 | 38.57 | 38.08 | 35.26 | 31.15 | 31.94 | 31.53 | 34.50 | 33.01 | 29.63 | 30.39 | 30.00 | 33.56 | 26.20 | 26.76 | 27.45 | 27.10 | 31.67 |
| UCAT (Ours) | 47.14 | 41.55 | 42.62 | 42.07 | 44.31 | **40.10** | **41.12** | **40.60** | 41.32 | 39.88 | **37.47** | **38.43** | **37.94** | **39.90** | 28.01 | **29.95** | **30.72** | **30.32** | **35.24** |